PARAMETER ESTIMATION IN NONLINEAR SYSTEMS WITH SATURATION NONLINEARITIES: A NOVEL APPROACH LEVERAGING STRUCTURAL PROPERTIES

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Abstract. We propose a method for parameter estimation in highly uncertain nonlinear systems whose measured output is the product of an unknown matrix with a known saturation function evaluated at the unmeasured state. Classical estimation techniques face significant difficulties in such settings, particularly when the system dynamics are only partially known. Our approach exploits the structural properties of the saturation itself: we construct explicit control inputs that drive the state into regions where the saturation takes predetermined values, without requiring knowledge of the state or of the underlying dynamics. This produces a signal issued from an unknown function of the state whose values are known at prescribed times, and whose variation over the time interval considered is informative enough to identify the unknown parameters. We give structural conditions under which the procedure applies, and extend the results to output feedback controllability under slightly stronger conditions. The main result is Theorem 2.3. Numerical simulations illustrate the method.

1. Introduction

The application of control and estimation methods to nonlinear systems is a fundamental challenge in many areas of science and engineering. Nonlinear phenomena such as saturations or more general activation functions are ubiquitous in real-world systems, and are particularly prevalent in neuroscience. Many widely used models of brain activity, such as the Wilson–Cowan model [10, 11] and Amari neural fields [1], explicitly incorporate such nonlinearities.

All dynamical models depend on a certain number of parameters whose values influence the behavior of the system and its response to controls. It is therefore critical to estimate, exactly or approximately, the value of these parameters. Some of them may be known from physical considerations, but most have to be estimated from indirect measurements, through *parameter identification* [7]. By identification we mean any procedure that uses measurements of the outputs of the system corresponding to some inputs, or excitation, to deduce an estimate of the parameters. The choice of the excitation, or experiment design, may be an important part of the process. Moreover, the state of the system may not be fully observed, and the estimation may have to rely on partial measurements; the problem is then one of state and parameter estimation, or adaptive observation.

These problems have been extensively studied since the foundational work in adaptive control and system identification [7, 3]. Classical approaches rely on systems with few parameters to estimate, and the standard framework assumes either contraction properties or passivity conditions ensuring convergence of the estimation errors. These assumptions are made to guarantee the recovery of the state independently of the unknown parameters. It is moreover usually assumed that the trajectories satisfy a persistent excitation condition, which has become the standard sufficient condition for parameter identifiability [3].

These frameworks rely on assumptions that do not hold generally. Many physical systems fail to satisfy global contraction or passivity properties, and a persistent excitation requirement bearing on unmeasured states is difficult to verify. Standard methods also require the unknown parameters to multiply either a known persistent signal, the regressor, or a state-dependent function assumed persistently excited through that same hypothesis. More and more models of complex systems, in neuroscience in particular, exhibit high uncertainty, with many unknown terms; often only the structure of the model is known, and the classical assumptions are not met.

This paper considers a class of highly uncertain nonlinear systems whose dynamics are unknown and whose output contains an unknown matrix applied to the saturation of the state. The problem is difficult because the unknown parameters do not multiply a known signal, and because the system satisfies none of the classical assumptions for estimation [3]. We propose to leverage the structure imposed by the saturation, and design explicit control inputs steering the state into regions where the saturation takes known values at prescribed times. The procedure is sequential. A bias and a linearly parametrized term, when present, are recovered first by regression on measured data over a window where the saturation vanishes. The unknown matrix is then identified by driving the state through distinct saturation regions and exploiting the resulting input–output relation. Finally, once the parameters are known, the state itself is recovered by output feedback control combined with algebraic inversion of the saturation. Each step comes with an explicit control design and an explicit estimation guarantee.

2. Main results

2.1. A general identification result

In this section we consider a system whose measured output is a linear function of the saturated state,

$$\begin{cases} \dot{x} = f(x,u,t), \\ y = WS(x), \end{cases} \tag{2.1}$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}^m$ the control, $y \in \mathbb{R}^q$ the measured output, $W \in \mathbb{R}^{q\times n}$ an unknown constant matrix and $S : \mathbb{R}^n \to \mathbb{R}^n$ a known continuous function. The vector field f is unknown; we only assume the regularity needed for the solutions to exist and to be absolutely continuous on the time interval under consideration. Nothing else about f is used in this section.

The problem addressed here is the identification of W from the measurement of y on a bounded time interval. The unknown parameters multiply the signal $S(x(\cdot))$, which is neither measured nor known and which is produced by a dynamics that is itself unknown. Classical regression is therefore not directly available, and neither is an observer, since no dissipativity or observability property of f is assumed.

The output in (2.1) is written without a bias and without a linearly parametrized term. Both can be added back, and this is done at the end of the section.

Throughout, $\|\cdot\|$ is the Euclidean norm on $\mathbb{R}^n$ and the matrix norm it induces, $\mathcal{B}(v,r)$ is the closed ball of centre v and radius r, so that the letter B remains available for the input matrix, and $\mathring{\Omega}$ is the interior of a set $\Omega \subset \mathbb{R}^n$. For two integers $p < q$ we write $[\![p,q]\!] = \{i \in \mathbb{Z} : \ p \le i \le q\}$.

The structural feature of S that we exploit is that it takes a fixed known value on large parts of the state space. This is what the following definition retains, and it is the only property of S used in this section.

Definition 2.1 (Saturation function). *A continuous function $S : \mathbb{R}^n \to \mathbb{R}^n$ is a* saturation function *if there exist a family of closed sets $\Sigma_i \subset \mathbb{R}^n$ with nonempty interior, i in some index set I, and vectors $v_i \in \mathbb{R}^n$, such that*

$$S(x) = v_i, \qquad \forall x \in \Sigma_i, \ \forall i \in I. \tag{2.2}$$

The set Σ_i is a saturation region *of S and v_i is its* saturation value. *The set of saturation values is denoted*

$$\mathcal{A}(S) := \{v_i : \; i \in I\}. \tag{2.3}$$

Nothing is required of S outside the saturation regions, and no relation is imposed between the regions themselves. The value v_i is known even though the state producing it is not, and this is the whole content of (2.2).

We now state the property on which the identification rests. It involves a trajectory, a finite collection of values and a finite collection of time intervals, and it does not involve the dynamics.

Definition 2.2 (Saturation-excitation condition). *An* excitation design *is a collection $\mathcal{E} = \{(v_i, J_i) \mid i = 1, \dots, N\}$ where the $J_i \subset \mathbb{R}$ are pairwise disjoint bounded intervals of positive length and the vectors $v_i \in \mathbb{R}^n$ satisfy*

$$\operatorname{rank}[v_1, \dots, v_N] = n. \tag{2.4}$$

An absolutely continuous trajectory $x : [0,T] \to \mathbb{R}^n$ is a realization of $\mathcal{E}$ *if $J_i \subset [0,T]$ for every i and*

$$S(x(t)) = v_i \qquad \text{for all } t \in J_i \text{ and all } i \in [\![1, N]\!]. \tag{2.5}$$

We say that x is saturation-exciting *if it realizes an excitation design.*

Condition (2.5) does not constrain the trajectory beyond the value taken by S along it, and in practice it is met by having $x(t)$ enter a saturation region Σ_i on which S equals v_i and stay there during J_i. It involves neither the value of x nor that of W, and the design $\mathcal{E}$ is fixed beforehand, independently of the solution. In particular (2.5) may hold for a trajectory which is not known, which is the situation of interest here: what has to be known is which values are produced and on which intervals, not where the state is inside the corresponding regions. Accordingly, (2.5) is established through a sufficient condition on the data of the problem and on the control, and not by inspection of x. Subsection 2.2 provides such conditions, under which every solution issued from a given ball realizes a prescribed design, uniformly with respect to the unknown part of the system.

One necessary condition can be read on the output. Under (2.5) the output verifies $y \equiv W v_i$ on J_i, and is therefore constant on each J_i even though W is unknown. The converse fails, since a trajectory may be at rest without being saturated.

Given an excitation design $\mathcal{E}$, write $\tau_i := |J_i|$ for the length of the i-th interval and define the excitation Gramian and the estimator

$$\Gamma(\mathcal{E}) := \sum_{i=1}^{N} \tau_i \, v_i v_i^\top \in \mathbb{R}^{n \times n}, \qquad E(y; \mathcal{E}) := \left(\sum_{i=1}^{N} \int_{J_i} y(s) \, v_i^\top \, ds \right) \Gamma(\mathcal{E})^{-1} \in \mathbb{R}^{q \times n}. \tag{2.6}$$

Theorem 2.3. *Let $\mathcal{E}$ be an excitation design and let (x, u, y) be a solution of (2.1) on $[0,T]$ whose state realizes $\mathcal{E}$. Then $\Gamma(\mathcal{E})$ is invertible and*

$$E(y; \mathcal{E}) = W. \tag{2.7}$$

Proof. Let $V := [v_1, \dots, v_N] \in \mathbb{R}^{n \times N}$ and $D := \operatorname{diag}(\tau_1, \dots, \tau_N)$, so that $\Gamma(\mathcal{E}) = V D V^\top$. For $z \in \mathbb{R}^n$ we have $z^\top \Gamma(\mathcal{E}) z = \sum_i \tau_i (v_i^\top z)^2$, which vanishes only if z is orthogonal to every v_i, hence only if $z = 0$ by (2.4). Thus $\Gamma(\mathcal{E})$ is positive definite.

By (2.5) the output verifies $y(t) = WS(x(t)) = Wv_i$ for every $t \in J_i$. Therefore

$$\sum_{i=1}^{N} \int_{J_i} y(s) v_i^\top \, ds = \sum_{i=1}^{N} \tau_i \, W v_i v_i^\top = W \, \Gamma(\mathcal{E}), \tag{2.8}$$

and multiplying on the right by $\Gamma(\mathcal{E})^{-1}$ gives the result. ◻

The estimator is explicit and non-asymptotic: it returns W at the end of the last interval and not in the limit. It uses no knowledge of f and no state observer, and it requires no rank assumption on W, the identifiability coming from the saturation and not from the output matrix. A recursive least-squares implementation of the same regression [7] is immediate.

Remark 2.4 (Designs of insufficient rank). Suppose the values of the design span only a proper subspace $V := \operatorname{span}\{v_1, \ldots, v_N\}$ of $\mathbb{R}^n$, so that (2.4) fails. This happens when S takes fewer than n independent values, and also when the values the dynamics can be made to produce do not span. The Gramian $\Gamma(\mathcal{E})$ is then singular with range V and Theorem 2.3 does not apply, but the data still determine part of W, and one can say exactly which. Writing $\Gamma(\mathcal{E})^+$ for the Moore–Penrose pseudoinverse and P_V for the orthogonal projector onto V, the computation of the proof is unchanged up to its last line and gives

$$\left(\sum_{i=1}^{N} \int_{J_i} y(s) \, v_i^\top \, ds \right) \Gamma(\mathcal{E})^+ \;=\; W \, \Gamma(\mathcal{E}) \, \Gamma(\mathcal{E})^+ \;=\; W P_V, \tag{2.9}$$

since $\Gamma(\mathcal{E})$ is symmetric with range V. The restriction of W to V is then recovered exactly.

Remark 2.5. Theorem 2.3 is stated for $S : \mathbb{R}^n \to \mathbb{R}^n$, but the equality of the two dimensions is not used in its proof. If $S : \mathbb{R}^n \to \mathbb{R}^d$ and $W \in \mathbb{R}^{q \times d}$, it suffices that the saturation values of the visited regions span $\mathbb{R}^d$, hence that $N \geq d$. This covers in particular the case where the saturation affects only part of the coordinates.

Remark 2.6 (Bias and linearly parametrized terms). Consider the more general output

$$y = WS(x) + h_0 + G(y, u, t)^\top \theta, \tag{2.10}$$

where $h_0 \in \mathbb{R}^q$ and $\theta \in \mathbb{R}^{n_\theta}$ are unknown and $G : \mathbb{R}^q \times \mathbb{R}^m \times \mathbb{R} \to \mathbb{R}^{n_\theta \times q}$ is known. Suppose that, besides the design $\mathcal{E}$, the trajectory visits during an interval J_0 preceding J_1 a region on which S vanishes, that is $S(x(t)) = 0$ for $t \in J_0$. On J_0 the output reduces to $y = \Phi\zeta$, with augmented regressor $\Phi := [\, I_q \quad G(y, u, \cdot)^\top] \in \mathbb{R}^{q \times (q + n_\theta)}$ and $\zeta := (h_0, \theta) \in \mathbb{R}^{q + n_\theta}$, a regression which is linear in the unknowns and which the least-squares estimator

$$E_0(y) := \left(\int_{J_0} \Phi(s)^\top \Phi(s) \, ds \right)^{-1} \int_{J_0} \Phi(s)^\top y(s) \, ds \tag{2.11}$$

inverts exactly as soon as the Gramian $\int_{J_0} \Phi^\top \Phi \, ds \in \mathbb{R}^{(q+n_\theta) \times (q+n_\theta)}$ is invertible. That requirement, unlike the persistent excitation of the classical theory, bears on a regressor which is entirely available, G being known and y and u measured, and can therefore be tested on the recorded data. Applying then Theorem 2.3 to $\tilde{y} := y - h_0 - G(y, u, \cdot)^\top \theta$ on $J_1, \ldots, J_N$ returns W. The two stages are sequential and acyclic: the second uses the output of the first, and the first uses nothing of W. The whole cost of the generalization is one additional region, the one carrying the value 0, which in the component-wise case is the requirement that the negative orthant be reachable.

Remark 2.7. The design $\mathcal{E}$ plays the role that the persistent excitation of the regressor plays in the classical theory, and the Gramian $\Gamma(\mathcal{E})$ that of the regressor Gramian. The difference is that here the regressor is $S(x(\cdot))$, which is neither measured nor known, so that no condition bearing on it could be checked; conditions weaker than persistence, such as excitation on a finite interval [6], initial excitation [8], concurrent learning [4] or dynamic regressor extension [2], all bear on the regressor as well and are of no help for that reason. What Definition 2.2 requires instead is that the regressor take N prescribed values, and the saturation is what makes those values known in advance. The excitation is designed rather than assumed.

2.2. Structural conditions achieving saturation control

Theorem 2.3 is of use only if (2.5) can be produced. We now specify the dynamics and give conditions under which an explicit control forces it. Let us consider the following realization of (2.1):

$$\begin{cases} \dot{x} = Ax + \phi(x, u, t) + Bu, \\ y = WS(x). \end{cases} \tag{2.12}$$

Here, we assume that A is Hurwitz and that there exists $\bar{\phi} > 0$ such that $\|\phi\| \leq \bar{\phi}$. The matrix $B \in \mathbb{R}^{n\times m}$ is known.

Three information structures are covered, and they differ only in what is known of the linear part. When A is known, the condition is a geometric one relating $\operatorname{Im}(A^{-1}B)$ to the saturation regions, and a controllability assumption on (A, B) removes the transient. When only a part A_0 of A is known, the same geometric condition is imposed on $\operatorname{Im}(A_0^{-1}B)$ and the unknown remainder is required to be small. When in addition the known part is a multiple of the identity, the transient can again be dispensed with.

The regions in which the state has to be brought are required to be unbounded in a controllable direction, which is what the following assumption expresses.

Assumption 2.8 (Coradiant saturation). *The function $S : \mathbb{R}^n \to \mathbb{R}^n$ is continuous. There exist n closed coradiant sets*[1] *$K_1, \dots, K_n \subset \mathbb{R}^n$ with non-empty interior, linearly independent vectors $v_1, \dots, v_n \in \mathbb{R}^n$, and $R > 0$ such that*

$$S(x) = v_i, \quad \forall x \in K_i,\ \|x\| \geq R, \quad i = 1, \dots, n. \tag{2.13}$$

Remark 2.9 (Classical saturations). The component-wise saturation is the model case. Let $S(x) = (S_0(x_1), \dots, S_0(x_n))^\top$ with $S_0 : \mathbb{R} \to \mathbb{R}$ nondecreasing, $S_0 = 0$ on $(-\infty, \underline{R}]$ and $S_0 = 1$ on $[\bar{R}, +\infty)$, put $R := \max\{|\underline{R}|, |\bar{R}|\}$ and, for a sign pattern $\mathbf{s} \in \{0, 1\}^n$, let $\mathcal{O}_{\mathbf{s}}$ be the closed orthant carrying that pattern and

$$K_{\mathbf{s}} := \big\{x \in \mathcal{O}_{\mathbf{s}} :\ |x_j| \geq R,\ j \in [\![1, n]\!]\big\}. \tag{2.14}$$

This set is closed, has nonempty interior, and is coradiant, since multiplying by $\lambda \geq 1$ preserves both the signs and the inequalities $|x_j| \geq R$; and $S = \mathbf{s}$ on the whole of $K_{\mathbf{s}}$,

[1] Recall that a set $K \subset \mathbb{R}^n$ is coradiant if $\lambda x \in K$ for all $x \in K$ and $\lambda \geq 1$.

because $s_j = 1$ forces $x_j \geq R \geq \bar{R}$ and $s_j = 0$ forces $x_j \leq -R \leq \underline{R}$. Assumption 2.8 therefore holds with n such sets and with $v_i \in \{0,1\}^n$, provided the corresponding patterns are linearly independent.

For $b \in \mathrm{Im}(B)$ and a Hurwitz A, a constant control $Bu = \mu b$ drives the state towards the steady state $-\mu A^{-1} b$. It is therefore the image of $\mathrm{Im}(B)$ under $-A^{-1}$, and not the image of B itself, which has to meet the regions. We have the following.

Theorem 2.10. *Assume that*

$$\mathrm{Im}(A^{-1}B) \cap \mathring{K}_i \neq \varnothing \qquad \forall i \in \{1, \dots, n\}$$

Then, there exists $T^* \geq 0$ such that for any $\rho > 0$ and $T > T^*$ there exists a control $u : [0, T] \to \mathbb{R}^m$ such that any associated solution x of (2.12) with $x(0) \in \mathcal{B}(0, \rho)$ is saturation-exciting.

The above theorem requires the knowledge of A. If this is not completely known we can still say something, as follows.

Theorem 2.11. *Assume that $A = A_0 + A_1$ with $\|A_1\| \leq \varepsilon_0$, where $\varepsilon_0 > 0$ depends only on A_0, B and $K_1, \dots, K_n$ and is given by (3.52), A_0 Hurwitz and such that*

$$\mathrm{Im}(A_0^{-1}B) \cap \mathring{K}_i \neq \varnothing \qquad \forall i \in \{1, \dots, n\}$$

Then, there exists $T^* > 0$ such that for any $\rho > 0$ and $T > T^*$ there exists a control $u : [0, T] \to \mathbb{R}^m$, computed from A_0, B, $K_1, \dots, K_n$, $\bar{\phi}$, ρ and the bound ε_0 alone, such that any associated solution x of (2.12) with $x(0) \in \mathcal{B}(0, \rho)$ is saturation-exciting.

Under additional assumptions on the unperturbed part A_0, we can actually do everything in arbitrary time.

Theorem 2.12. *Under the assumptions of Theorem 2.11, assume that either*

i. $A_1 = 0$ and the couple (A_0, B) is controllable;

ii. $A_0 = -\lambda\,\mathrm{Id}$ for some $\lambda > 0$.

Then, for any $\rho > 0$ and $T > 0$ there exists a control $u : [0, T] \to \mathbb{R}^m$, computed from the same data, such that any associated solution x of (2.12) with $x(0) \in \mathcal{B}(0, \rho)$ is saturation-exciting.

The time T^* of Theorems 2.10 and 2.11 is the total duration of the transients that bring the state into the successive regions; the dwell times spent inside them may on the other hand be taken arbitrarily short, since they are part of the design and not of the dynamics.

2.3. Recovering the state via output feedback control

Once W has been identified, the output carries information about the state on the part of the space where S has not saturated. This suggests the control problem complementary to the previous one: instead of steering the state into the saturation regions, keep it out of them. Doing so, and reading the state off the output afterwards, needs more of S than Assumption 2.8 provides: that assumption says nothing outside the saturation regions, and in particular does not make S injective anywhere. What is needed is that the output be, after a known change of variables, a component-wise saturation of the state.

Assumption 2.13 (Component-wise invertible output). *There exist a matrix $H \in \mathbb{R}^{n \times q}$, levels $\underline{R} < \bar{R}$ and a continuous nondecreasing $S_0 : \mathbb{R} \to [0, 1]$, equal to 0 on $(-\infty, \underline{R}]$, equal to 1 on $[\bar{R}, +\infty)$ and strictly increasing on $[\underline{R}, \bar{R}]$, such that*

$$H\,W\,S(x) \;=\; S_1(x) := \big(S_0(x_1), \dots, S_0(x_n)\big)^\top, \qquad \forall x \in \mathbb{R}^n. \tag{2.15}$$

The signal $Hy = S_1(x)$ is then computable from the output, and it is a component-wise saturation of the state, on which the scalar feedback used below can act coordinate by coordinate. We write

$$\Omega := [\underline{R}, \bar{R}]^n \tag{2.16}$$

for the region on which S_1 has not saturated; there S_0 is strictly increasing, hence invertible, and that is the only invertibility the recovery uses.

Remark 2.14. The model case is S itself component-wise, as in Remark 2.9, together with $\operatorname{rank} W = n$: then W admits a left inverse $W^\dagger$ and $H = W^\dagger$ gives (2.15) with $S_1 = S$. More generally, if $S = PS_1$ for a known invertible $P \in \mathbb{R}^{n\times n}$ and a component-wise S_1, one takes $H = P^{-1}W^\dagger$. Conversely (2.15) forces $\operatorname{rank} W = n$, hence $q \geq n$: the values of S_1 span $\mathbb{R}^n$, so HW is onto and $\operatorname{rank} W \geq n$. Note that the identification of W by Theorem 2.3 required no rank assumption at all; the rank enters only when the state itself is to be recovered.

Theorem 2.15. *Assume that B is invertible, that W is known and that Assumption 2.13 holds. Let $\rho > 0$ and $\varepsilon > 0$, let $0 < T_1 < T_2$, and let γ be a C^1 curve with bounded derivative taking its values in a compact subset of $\mathring{\Omega}$. Then there exists an explicit output feedback law η, given by (4.3), such that every solution of (2.12) under the control $u(t) = \eta(y(t), \gamma(t))$ with $x(0) \in \mathcal{B}(0, \rho)$ is global in time and satisfies*

$$x(t) \in \mathring{\Omega} \quad \forall t \geq T_1, \qquad \|x(t) - \gamma(t)\| \leq \varepsilon \quad \forall t \geq T_2. \tag{2.17}$$

The reference γ is at the user's disposal and may be taken constant. The first assertion is what matters here: the state is brought out of the saturation regions, and kept out of them from a prescribed time on. The second says that it can moreover be placed anywhere inside the unsaturated region, and not merely kept there. Once the state is unsaturated, the output determines it.

Corollary 2.16 (State recovery). *Under the assumptions of Theorem 2.15, the restriction of S_1 to Ω is a homeomorphism onto $[0, 1]^n$, and for every $t \geq T_1$*

$$x_j(t) = S_0^{-1}\Big(\big(Hy(t)\big)_j\Big), \qquad j \in [\![1, n]\!]. \tag{2.18}$$

The three results compose. Theorem 2.11 produces a trajectory realizing an excitation design, Theorem 2.3 reads W off the corresponding output, and the feedback of Theorem 2.15 then brings the state into the unsaturated region, where (2.18) returns it, or tracks a prescribed reference inside that region.

3. Steering into the saturation regions

This section proves Theorems 2.10, 2.11 and 2.12. Each of them is obtained by steering the state into one region K_i at a time and concatenating the resulting controls, and the four steering lemmas of Subsection 3.1 differ only in the information available on the linear part. We label them R1 (A known), R1′ (A known with (A, B) controllable), R2 (A_0 and a bound on $\|A_1\|$ known) and R3 (the case $A_0 = -\lambda \operatorname{Id}$ of R2), and refer to them by name below. Only R2 and R3 leave part of the linear dynamics unknown; only R1 and R2 need a minimal time.

Theorem 2.10 uses R1, or R1′ when the pair is controllable; Theorem 2.11 uses R2; the two cases of Theorem 2.12 use R1′ and R3. Theorem 2.15 and Corollary 2.16 rest on a different mechanism and are proved in Section 4.

Throughout, $M \geq 1$ and $\alpha > 0$ denote constants with

$$\|e^{At}\| \leq M e^{-\alpha t}, \qquad t \geq 0, \tag{3.1}$$

which exist because A is Hurwitz, and M_0, α_0 denote the analogous constants for A_0 when the decomposition $A = A_0 + A_1$ of Theorems 2.11 and 2.12 is in force. Integrating (3.1) over $[0, +\infty)$ gives $\|A^{-1}\| \leq M/\alpha$, and likewise $\|A_0^{-1}\| \leq M_0/\alpha_0$. The field ϕ is a Carathéodory function and the controls are bounded and measurable, so that solutions of (2.12) exist in the sense of Carathéodory, are absolutely continuous, and are global as soon as they are bounded. Uniqueness is neither assumed nor used: every conclusion is asserted for every solution issued from the prescribed set of initial states.

The geometry enters through a single scalar, which measures how deep inside a region a direction points.

Definition 3.1 (Margin). *Let $K \subset \mathbb{R}^n$ be closed and coradiant with nonempty interior, and let $v \in \mathring{K}$. The* margin *of v in K is*

$$\delta_K(v) := \min\left\{1,\ \frac{\operatorname{dist}(v, \mathbb{R}^n \setminus K)}{\|v\|}\right\} \in (0, 1]. \tag{3.2}$$

When K is a cone, $\delta_K(v)$ is the sine of the smallest angle between v and a direction leaving K; on an orthant and for v in its interior it is $\min_j |v_j|/\|v\|$, and on a convex cone the sine of the angle from v to the nearest face.

This depth is preserved under scaling: moving out along the ray $\mathbb{R}_+ v$ widens the admissible tube proportionally, and a perturbation of bounded size is thereby absorbed. Below, r stands for that perturbation.

Proposition 3.2 (Stable scaling of coradiant sets). *Let $K \subset \mathbb{R}^n$ be closed and coradiant with nonempty interior, let $v \in \mathring{K}$ and set $\delta := \delta_K(v) \in (0, 1]$. Then for every $\mu \geq 1$ and every $r \in \mathbb{R}^n$,*

$$\operatorname{dist}\big(\mu v + r,\ \mathbb{R}^n \setminus K\big)\ \geq\ \mu\delta\|v\| - \|r\|. \tag{3.3}$$

In particular, $\mu v + r \in K$ whenever $\|r\| \leq \mu\delta\|v\|$, and $\|\mu v + r\| \geq \mu\|v\| - \|r\|$.

Proof. By definition of δ we have $\mathcal{B}(v, \delta\|v\|) \subset K$. For $\mu \geq 1$, scaling a ball scales its centre and its radius, and coradiancy gives $\mu K \subset K$, so that

$$\mathcal{B}(\mu v,\ \mu\delta\|v\|) = \mu\, \mathcal{B}(v, \delta\|v\|) \subset \mu K \subset K. \tag{3.4}$$

If $\|r\| \leq \mu\delta\|v\|$ then $\mu v + r$ lies in that ball, hence in K. The distance estimate (3.3) follows from (3.4) and the triangle inequality

$$\operatorname{dist}\big(\mu v + r,\ \mathbb{R}^n \setminus K\big)\ \geq\ \operatorname{dist}\big(\mu v,\ \mathbb{R}^n \setminus K\big) - \|r\|,$$

and the norm bound is the reverse triangle inequality. ∎

The trajectories are read through Duhamel's formula: for a solution of (2.12),

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-s)}\big[\phi(s) + Bu(s)\big]\,ds, \qquad t \geq 0, \tag{3.5}$$

where $\phi(s)$ stands for $\phi(x(s), u(s), s)$. Fix a steering direction $b \in \operatorname{Im}(B)$ and take the constant control $Bu \equiv \mu b$ with $\mu > 0$. Since $\int_0^t e^{A(t-s)}ds = A^{-1}(e^{At} - \mathrm{Id})$, that control contributes $\mu(e^{At}-\mathrm{Id})A^{-1}b$ to (3.5), and grouping the constant part of this with the first term gives

$$x(t) = \underbrace{-\mu A^{-1}b}_{\text{equilibrium}} + \underbrace{e^{At}\big(x(0)+\mu A^{-1}b\big)}_{r_1(t)} + \underbrace{\int_0^t e^{A(t-s)}\phi(s)\,ds}_{r_2(t)}, \tag{3.6}$$

where, for $\|x(0)\| \le \rho$ and by (3.1) and $\|\phi\| \le \bar\phi$,

$$\|r_1(t)\| \;\le\; Me^{-\alpha t}\big(\rho+\mu\|A^{-1}b\|\big), \qquad \|r_2(t)\| \;\le\; \frac{M\bar\phi}{\alpha}, \qquad t \ge 0. \tag{3.7}$$

The amplitude scales the equilibrium and the transient r_1, but not r_2. Regrouping instead the two terms of (3.6) that carry μ,

$$x(t) = \mu\big(\mathrm{Id}-e^{At}\big)\big(-A^{-1}b\big) + \underbrace{e^{At}x(0) + \int_0^t e^{A(t-s)}\phi(s)\,ds}_{\text{of norm at most } M\rho+M\bar\phi/\alpha,\text{ uniformly in } t \text{ and in } \mu}, \tag{3.8}$$

puts the whole dependence on the amplitude in the curve $t \mapsto (\mathrm{Id}-e^{At})(-A^{-1}b)$. The first form is used when the transient is waited out, the second when it is not. In either case (3.7) gives the uniform bound

$$\|x(t)\| \;\le\; R_\mu := \mu(1+M)\big\|A^{-1}b\big\| + M\rho + \frac{M\bar\phi}{\alpha}, \qquad t \ge 0. \tag{3.9}$$

Each lemma of Subsection 3.1 establishes the following property, and only this property is used afterwards.

Definition 3.3 (Steering into a region). *Let $K \subset \mathbb{R}^n$ be closed and coradiant with nonempty interior and let $T_0 > 0$. We say that* (2.12) can be steered into K in time T_0 *if for every $\rho > 0$ there exist a bounded measurable control $u : [0,+\infty) \to \mathbb{R}^m$ and a radius $\rho' \ge \rho$, neither of them depending on the solution, such that every solution of* (2.12) *with $x(0) \in \mathcal{B}(0,\rho)$ satisfies*

$$x(t) \in K \;\; \textit{and} \;\; \|x(t)\| \ge R \quad \forall t \ge T_0, \qquad \|x(t)\| \le \rho' \quad \forall t \ge 0. \tag{3.10}$$

Under Assumption 2.8 and for $K = K_i$, (2.13) turns (3.10) into $S(x(t)) = v_i$ for every $t \ge T_0$, the value that (2.5) prescribes on J_i. The radius ρ' bounds the state at the end of the step and allows the regions to be visited one after the other.

3.1. Steering into a coradiant region

All the constants below depend on the steering direction only through its depth in the region, so we first record the best depth a region affords. For a Hurwitz matrix A' set

$$\delta_K^*(A') := \sup\big\{\delta_K(w) : \; w \in \mathrm{Im}(A'^{-1}B) \cap \mathring{K}\big\} \;\in\; [0,1], \tag{3.11}$$

with the convention $\sup\varnothing = 0$, so that $\delta_K^*(A') > 0$ is equivalent to $\mathrm{Im}(A'^{-1}B) \cap \mathring{K} \ne \varnothing$. Every $\eta < \delta_K^*(A')$ is then the depth of an admissible direction: there is $b \in \mathrm{Im}(B)$ with $-A'^{-1}b \in \mathring{K}$ and $\delta_K(-A'^{-1}b) > \eta$.

The supremum is in general not a maximum, and closedness of K does not help: the competitors are unbounded and the best of them escape to infinity. Indeed δ_K is nondecreasing along each ray, since $\mathcal{B}(v,d) \subset K$ and $t \geq 1$ give $\mathcal{B}(tv,td) = t\,\mathcal{B}(v,d) \subset tK \subset K$ by coradiancy, so that $\delta_K(tv) \geq \delta_K(v)$. For the component-wise saturations of Remark 2.9 the increase is strict and the limit $\min_j |w_j|/\|w\|$ is never attained. We therefore prove every statement below for a direction of a given depth η, and let the theorems of Subsection 3.3 take η close to δ_K^*.

LEMMA 3.4 (Steering lemma). *Let A be Hurwitz, let $K \subset \mathbb{R}^n$ be closed and coradiant with nonempty interior, let $\rho > 0$ and assume that*

$$\operatorname{Im}(A^{-1}B) \cap \mathring{K} \neq \varnothing. \tag{3.12}$$

Let $b \in \operatorname{Im}(B)$ be such that $-A^{-1}b$ belongs to that intersection, and set

$$\delta := \delta_K\big(-A^{-1}b\big), \qquad \chi(T_0) := \delta - Me^{-\alpha T_0}, \qquad T_{\min} := \frac{1}{\alpha}\ln\frac{M}{\delta}. \tag{3.13}$$

Then $\chi(T_0) > 0$ if and only if $T_0 > T_{\min}$, and for every such T_0 and every

$$\mu \;\geq\; \mu^\star := \max\left\{1,\ \frac{R + M\rho + M\bar{\phi}/\alpha}{\chi(T_0)\,\|A^{-1}b\|}\right\}, \tag{3.14}$$

every solution of (2.12) with $x(0) \in \mathcal{B}(0,\rho)$ driven by the constant control $Bu \equiv \mu b$ satisfies

$$x(t) \in K \quad \text{and} \quad \|x(t)\| \geq R \quad \forall t \geq T_0, \qquad \|x(t)\| \leq R_\mu \quad \forall t \geq 0. \tag{3.15}$$

This is (3.10) with $\rho' = R_\mu$, so (2.12) can be steered into K in every time $T_0 > T_{\min}$.

Proof. Write $w := -A^{-1}b$ for the equilibrium the control aims at. Since $b \in \operatorname{Im}(B)$ the constant control $Bu \equiv \mu b$ is admissible, and Duhamel's formula (3.5) together with $\int_0^t e^{A(t-s)}ds = A^{-1}(e^{At} - \mathrm{Id})$ gives, as in (3.6),

$$x(t) = \mu w + \underbrace{e^{At}\big(x(0) - \mu w\big)}_{r_1(t)} + \underbrace{\int_0^t e^{A(t-s)}\phi(s)\,ds}_{r_2(t)}, \qquad t \geq 0. \tag{3.16}$$

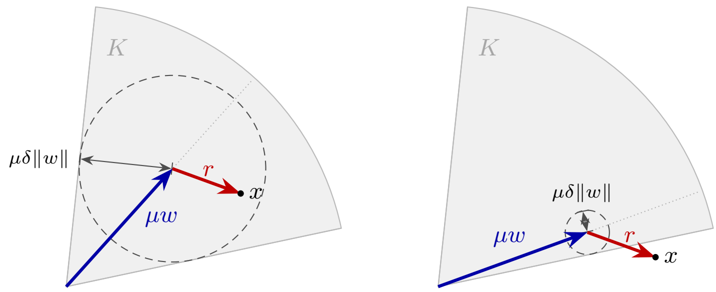
The state is thus μw up to a residual, and by (3.7), for $t \geq T_0$,

$$\|r_1(t) + r_2(t)\| \;\leq\; \underbrace{\mu M e^{-\alpha T_0}\|w\|}_{\text{grows with } \mu} + \underbrace{M\rho + \frac{M\bar{\phi}}{\alpha}}_{\text{does not}}, \tag{3.17}$$

where we used $Me^{-\alpha T_0}\rho \leq M\rho$. As $\mu \geq 1$, the tube estimate (3.3) applies to μw and gives $\operatorname{dist}(\mu w, \mathbb{R}^n \setminus K) \geq \mu\delta\|w\|$. The two terms of (3.17) are dominated separately: the first by the margin, since $Me^{-\alpha T_0} < \delta$ is the definition of $T_0 > T_{\min}$, and the second by the amplitude. Together, with $\chi(T_0) = \delta - Me^{-\alpha T_0}$ and (3.14),

$$\mu\delta\|w\| - \|r_1(t) + r_2(t)\| \;\geq\; \mu\chi(T_0)\|w\| - M\rho - \frac{M\bar{\phi}}{\alpha} \;\geq\; R \;>\; 0. \tag{3.18}$$

In particular $\|r_1(t) + r_2(t)\| \leq \mu\delta\|w\|$, so Proposition 3.2 applied with $v = w$ and $r = r_1(t) + r_2(t)$ gives $x(t) \in K$, and its norm bound together with (3.18) gives $\|x(t)\| \geq \mu\|w\| - \|r_1(t) + r_2(t)\| \geq R$. Both hold for every $t \geq T_0$, and (3.9) bounds the state on $[0, +\infty)$. $\square$



deep direction: δ large,
the room absorbs r

shallow direction: δ small,
the same r leaves K

$x = \mu w + r$

FIGURE 3.1. *The decomposition $x = \mu w + r$ of (3.16), with $w = -A^{-1}b$ the equilibrium the control aims at and r the residual. By Proposition 3.2 the region contains the ball of radius $\mu\delta\|w\|$ centered at μw (dashed), so x lies in K as soon as $\|r\| \leq \mu\delta\|w\|$. The depth δ of the direction fixes the radius of that ball: on the left it absorbs the residual, on the right it does not. The radius grows with the amplitude, so the part of r that stays bounded is always beaten by taking μ large. The part that grows with μ is not: the transient $\mu M e^{-\alpha T_0}\|w\|$ of (3.17), and, in Lemma 3.5, the displacement of the equilibrium caused by the unknown A_1. Those must be smaller than δ, whence the minimal time (3.13) and the condition (3.20).*

The minimal time comes from the first term of (3.17). The transient grows like μ, exactly as the margin does, so one has to wait for it to fall below the margin and no amplitude shortens that wait. The second term is the only one the amplitude has to beat. Figure 3.1 shows the two panels of this comparison.

We turn to the case where only a part A_0 of the linear dynamics is known, the control being built from A_0 while the equilibrium it reaches, $-A^{-1}b$, involves the unknown part. Write

$$(3.19) \qquad p := \left\|A_0^{-1}A_1\right\| \leq \frac{M_0}{\alpha_0}\|A_1\|, \qquad \theta := \frac{p}{1-p},$$

so that θ measures, relative to $\|A_0^{-1}b\|$, how far A_1 can move that equilibrium. The whole case is governed by one comparison: the displacement θ must be shorter than the depth of the nominal direction.

LEMMA 3.5 (Steering lemma, partially known linear part). *Let A_0 be Hurwitz with constants M_0, α_0 as in (3.1), let $K \subset \mathbb{R}^n$ be closed and coradiant with nonempty interior and let $\rho > 0$. Let $b \in \mathrm{Im}(B)$ be such that $w_0 := -A_0^{-1}b \in \mathring{K}$, and set $\delta_0 := \delta_K(w_0)$. Assume*

$$(3.20) \qquad \|A_1\| < \frac{\alpha_0}{M_0} \cdot \frac{\delta_0}{1+\delta_0}, \qquad \textit{which by (3.19) gives} \qquad \theta < \delta_0.$$

Then $A := A_0 + A_1$ is Hurwitz with $\|e^{At}\| \leq M_0 e^{-\alpha_0 t/2}$, and the equilibrium $w := -A^{-1}b$ satisfies

$$(3.21) \qquad w \in \mathring{K}, \qquad \delta_K(w) \geq \frac{\delta_0 - \theta}{1+\theta} > 0, \qquad \|w\| \geq (1-\theta)\|w_0\|.$$

Set $\chi_0(T_0) := (\delta_0 - \theta)/(1+\theta) - M_0 e^{-\alpha_0 T_0/2}$. Then $\chi_0(T_0) > 0$ if and only if

$$(3.22) \qquad T_0 > T^0_{\min} := \frac{2}{\alpha_0} \ln \frac{M_0(1+\theta)}{\delta_0 - \theta},$$

and for every such T_0 and every

$$\mu \;\geq\; \max\left\{1,\; \frac{R + M_0\rho + 2M_0\bar{\phi}/\alpha_0}{\chi_0(T_0)\,(1-\theta)\|w_0\|}\right\}, \tag{3.23}$$

every solution of (2.12) *with* $x(0) \in \mathcal{B}(0,\rho)$ *driven by the constant control* $Bu \equiv \mu b$ *satisfies*

$$x(t) \in K \quad \text{and} \quad \|x(t)\| \geq R \quad \forall t \geq T_0, \qquad \|x(t)\| \leq R_\mu \quad \forall t \geq 0. \tag{3.24}$$

This is (3.10) *with* $\rho' = R_\mu$, *so* (2.12) *can be steered into* K *in every time* $T_0 > T^0_{\min}$.

Every quantity in (3.20)–(3.23) is computed from A_0, B, K, ρ, $\bar{\phi}$ and an upper bound on $\|A_1\|$; the matrix A_1 itself is never used, and neither is the true margin $\delta_K(w)$, of which only the computable lower bound in (3.21) enters. All the constants are monotone in θ, so an upper bound may be substituted for it throughout.

Proof. Step 1: where the equilibrium goes. Write $P := A_0^{-1}A_1$, so that $A = A_0(\mathrm{Id} + P)$ with $\|P\| = p$. By (3.20) and $\delta_0 \leq 1$ we have $p < \delta_0/(1+\delta_0) \leq 1/2$, so the Neumann series inverts $\mathrm{Id} + P$, hence A, with $w = (\mathrm{Id} + P)^{-1}w_0$ and

$$\|w - w_0\| = \big\|(\mathrm{Id} + P)^{-1}Pw_0\big\| \;\leq\; \frac{p}{1-p}\|w_0\| \;=\; \theta\|w_0\|, \tag{3.25}$$

whence $(1-\theta)\|w_0\| \;\leq\; \|w\| \;\leq\; (1+\theta)\|w_0\|$ by the triangle inequality. Since $p < \delta_0/(1+\delta_0)$ is exactly $\theta < \delta_0$, the displacement (3.25) is shorter than the radius $\delta_0\|w_0\|$ of the ball that K contains around w_0, so

$$\mathrm{dist}\big(w, \mathbb{R}^n \setminus K\big) \;\geq\; (\delta_0 - \theta)\|w_0\| \;>\; 0,$$

which places w in $\mathring{K}$; dividing by $\|w\| \leq (1+\theta)\|w_0\|$ and using $(\delta_0 - \theta)/(1+\theta) \leq 1$ gives (3.21).

Step 2: the decay of e^{At}. From $e^{At} = e^{A_0 t} + \int_0^t e^{A_0(t-s)}A_1 e^{As}ds$ and (3.1) for A_0, the function $\psi(t) := e^{\alpha_0 t}\|e^{At}\|$ satisfies $\psi(t) \leq M_0 + M_0\|A_1\|\int_0^t \psi(s)ds$, so Grönwall's lemma gives $\|e^{At}\| \leq M_0 e^{-(\alpha_0 - M_0\|A_1\|)t}$. Here $M_0\|A_1\| < \alpha_0\delta_0/(1+\delta_0) \leq \alpha_0/2$, so (3.1) holds with $M = M_0$ and $\alpha > \alpha_0/2$, and in particular A is Hurwitz and $M\bar{\phi}/\alpha < 2M_0\bar{\phi}/\alpha_0$.

Step 3: Duhamel along the perturbed equilibrium. The control $Bu \equiv \mu b$ is admissible and constant, so (3.5) and $\int_0^t e^{A(t-s)}ds = A^{-1}(e^{At} - \mathrm{Id})$ give, exactly as in (3.16) but with the equilibrium w of Step 1,

$$x(t) = \mu w + \underbrace{e^{At}\big(x(0) - \mu w\big)}_{r_1(t)} + \underbrace{\int_0^t e^{A(t-s)}\phi(s)\,ds}_{r_2(t)}, \qquad t \geq 0, \tag{3.26}$$

and Step 2 bounds the residual, for $t \geq T_0$, by

$$\|r_1(t) + r_2(t)\| \;\leq\; \mu M_0 e^{-\alpha_0 T_0/2}\|w\| + M_0\rho + \frac{2M_0\bar{\phi}}{\alpha_0}. \tag{3.27}$$

As $\mu \geq 1$, (3.3) gives $\mathrm{dist}(\mu w, \mathbb{R}^n \setminus K) \geq \mu\delta_K(w)\|w\|$. Subtracting (3.27) and using (3.21) twice, first to bound $\delta_K(w) - M_0 e^{-\alpha_0 T_0/2}$ from below by $\chi_0(T_0)$ and then to bound $\|w\|$ from below by $(1-\theta)\|w_0\|$,

$$\mu\delta_K(w)\|w\| - \|r_1(t) + r_2(t)\| \;\geq\; \mu\chi_0(T_0)(1-\theta)\|w_0\| - M_0\rho - \frac{2M_0\bar{\phi}}{\alpha_0} \;\geq\; R \;>\; 0 \tag{3.28}$$

by (3.23). Proposition 3.2, applied with $v = w$ and $r = r_1(t) + r_2(t)$, then gives $x(t) \in K$, and its norm bound with (3.28) gives $\|x(t)\| \geq R$, for every $t \geq T_0$. The bound (3.9) holds on $[0, +\infty)$, which is (3.24). □

Condition (3.20) compares a displacement with a depth, both measured relative to $\|w_0\|$: the unknown part may move the equilibrium anywhere inside the ball that K already contains around the nominal one, and no further. It is the right panel of Figure 3.1, with $\theta\|w_0\|$ in place of $\|r\|$, and no amplitude repairs a violation of it: by (3.26) the state is μw up to an error that does not grow with μ, so once w has left K, raising the amplitude drives the state further from the region. Since the hypothesis is monotone in δ_0, a direction realizing it exists as soon as

$$\|A_1\| \;<\; \frac{\alpha_0}{M_0} \cdot \frac{\delta_K^*(A_0)}{1 + \delta_K^*(A_0)}, \tag{3.29}$$

a condition on the data alone: it suffices to take δ_0 close enough to $\delta_K^*(A_0)$ in (3.11).

When the known part is a multiple of the identity, the transient no longer has to be waited out. What must then stay inside K is the curve $t \mapsto (\operatorname{Id} - e^{At})(-A^{-1}b)$ of (3.8), and that curve stays on the ray of its own limit up to a controlled error at every time, not only asymptotically. The price is a factor two in the condition on A_1.

Lemma 3.6 (Steering lemma in arbitrary time, scalar known part). *Let $A_0 = -\lambda \operatorname{Id}$ with $\lambda > 0$, let $K \subset \mathbb{R}^n$ be closed and coradiant with nonempty interior and let $\rho > 0$; here $M_0 = 1$, $\alpha_0 = \lambda$, $w_0 = -A_0^{-1}b = b/\lambda$ and $p = \|A_1\|/\lambda$ in (3.19). Let $b \in \operatorname{Im}(B)$ be such that $b/\lambda \in \mathring{K}$ and assume*

$$2\theta \;<\; \delta_0 := \delta_K(b/\lambda). \tag{3.30}$$

Then, for every $T_0 > 0$ and every

$$\mu \;\geq\; \max\left\{ \frac{1}{1 - e^{-\lambda T_0}},\; \frac{\lambda(1+\theta)\big(R + \rho + 3\bar{\phi}/(2\lambda)\big)}{\big(1 - e^{-\lambda T_0}\big)(1-\theta)(\delta_0 - 2\theta)\,\|b\|} \right\}, \tag{3.31}$$

every solution of (2.12) with $x(0) \in \mathcal{B}(0, \rho)$ driven by the constant control $Bu \equiv \mu b$ satisfies

$$x(t) \in K \quad \text{and} \quad \|x(t)\| \geq R \quad \forall t \geq T_0, \qquad \|x(t)\| \leq R_\mu \quad \forall t \geq 0. \tag{3.32}$$

This is (3.10) with $\rho' = R_\mu$, so (2.12) can be steered into K in every time $T_0 > 0$.

Proof. Since $2\theta < \delta_0 \leq 1$ we have $\theta < \delta_0/2$, hence $p = \theta/(1+\theta) < \delta_0/(2+\delta_0) < \delta_0/(1+\delta_0)$, and $p = \|A_1\|/\lambda$ makes this the hypothesis (3.20) of Lemma 3.5. Its conclusion (3.21) places $w := -A^{-1}b$ in $\mathring{K}$, with

$$\hat{\delta} := \delta_K(w) \;\geq\; \frac{\delta_0 - \theta}{1 + \theta}, \qquad \|w\| \;\geq\; (1-\theta)\frac{\|b\|}{\lambda}. \tag{3.33}$$

Moreover $\theta < 1/2$ gives $p < 1/3$, hence $\alpha = \lambda(1-p) > 2\lambda/3$ below.

Since $-\lambda \operatorname{Id}$ commutes with A_1 we have $e^{At} = e^{-\lambda t} e^{A_1 t}$, so $\|e^{At}\| \leq e^{-(\lambda - \|A_1\|)t}$ and $M = 1$ may be taken in (3.1). Under the constant control $Bu \equiv \mu b$, Duhamel's formula (3.5) written in the form (3.8), which keeps the whole amplitude in a single term, reads

$$x(t) = \mu\big(\operatorname{Id} - e^{At}\big)w + e(t), \qquad e(t) := e^{At}x(0) + \int_0^t e^{A(t-s)}\phi(s)\,ds, \qquad \|e(t)\| \leq \rho + \frac{\bar{\phi}}{\alpha}, \tag{3.34}$$

the residual e being bounded uniformly in t and in μ. It therefore suffices to keep the curve $t \mapsto (\mathrm{Id} - e^{At})w$ deep enough inside K for every $t \geq T_0$. Writing $\sigma_t := 1 - e^{-\lambda t}$,

$$(3.35) \qquad \big(\mathrm{Id} - e^{At}\big)w = \sigma_t\, w + z(t), \qquad z(t) := e^{-\lambda t}\big(\mathrm{Id} - e^{A_1 t}\big)w.$$

The first term lies on the ray $\mathbb{R}_+ w$ for every $t > 0$, which is the difference with the general case, where the transient also rotates the state away from that ray; the whole shape error is thus carried by z. From $\|\mathrm{Id} - e^{A_1 t}\| \leq e^{\|A_1\| t} - 1$,

$$(3.36) \qquad \|z(t)\| \;\leq\; e^{-\lambda t}\big(e^{\|A_1\| t} - 1\big)\|w\| \;=\; \sigma_t\, \frac{e^{-\lambda t}\big(e^{\|A_1\| t} - 1\big)}{1 - e^{-\lambda t}}\, \|w\|,$$

and we claim that the fraction is bounded by $\|A_1\|/\lambda$ for every $t > 0$. Setting $s := e^{-\lambda t} \in (0,1)$ and $\beta := 1 - \|A_1\|/\lambda \in (0,1)$, it reads $(s^\beta - s)/(1-s)$, and the weighted arithmetic–geometric mean inequality $s^\beta = s^\beta \cdot 1^{1-\beta} \leq \beta s + (1-\beta)$ is the assertion $(s^\beta - s)/(1-s) \leq 1 - \beta = \|A_1\|/\lambda$. Hence

$$(3.37) \qquad \|z(t)\| \;\leq\; \sigma_t\, \frac{\|A_1\|}{\lambda}\, \|w\|, \qquad t > 0.$$

Both terms of (3.35) carry the same factor σ_t, so their comparison is independent of t and of μ. The minimal time disappears for that reason: the shape error need not decay, being proportional to the term it is compared with.

Let now $\mu \geq \sigma_{T_0}^{-1}$ and $t \geq T_0$, so that $\mu\sigma_t \geq \mu\sigma_{T_0} \geq 1$ and (3.3) applies to $\mu\sigma_t w$, giving $\mathrm{dist}(\mu\sigma_t w, \mathbb{R}^n \setminus K) \geq \mu\sigma_t\hat{\delta}\|w\|$. Subtracting $\mu\|z(t)\|$ and using (3.37) and the monotonicity of σ_t,

$$(3.38) \qquad \mathrm{dist}\Big(\mu\big(\mathrm{Id} - e^{At}\big)w,\ \mathbb{R}^n \setminus K\Big) \;\geq\; \mu\sigma_{T_0}\|w\| \left(\hat{\delta} - \frac{\|A_1\|}{\lambda}\right),$$

while the same two bounds give $\mu\|(\mathrm{Id} - e^{At})w\| \;\geq\; \mu\sigma_{T_0}\|w\|(1 - \|A_1\|/\lambda)$, a larger quantity since $\hat{\delta} \leq 1$. By (3.34), (3.32) holds as soon as the right-hand side of (3.38) is at least $R + \rho + \bar{\phi}/\alpha$. Three bounds turn this into (3.31): first $\|A_1\|/\lambda = \theta/(1+\theta)$ and (3.33) give

$$(3.39) \qquad \hat{\delta} - \frac{\|A_1\|}{\lambda} \;\geq\; \frac{\delta_0 - \theta}{1+\theta} - \frac{\theta}{1+\theta} \;=\; \frac{\delta_0 - 2\theta}{1+\theta} \;>\; 0,$$

which is where the factor two enters; then $\|w\| \geq (1-\theta)\|b\|/\lambda$; then $\bar{\phi}/\alpha < 3\bar{\phi}/(2\lambda)$. The requirement $\mu \geq \sigma_{T_0}^{-1}$ is the first term of the maximum in (3.31). ◻

As in Lemma 3.5, a suitable direction exists as soon as A_1 obeys a single bound on the data, here

$$(3.40) \qquad \|A_1\| \;<\; \lambda \cdot \frac{\delta_K^*(A_0)}{2 + \delta_K^*(A_0)},$$

which is equivalent to $2\theta < \delta_K^*(A_0)$; a direction of depth $\delta_0 > 2\theta$ then exists by (3.11).

Condition (3.40) does not involve T_0. No minimal time is needed, the price being paid on the amplitude, which by (3.31) grows like $1/(\lambda T_0)$ as the prescribed time shrinks. The condition is also of a different nature from (3.20): it bounds the rotation of the curve $t \mapsto (\mathrm{Id} - e^{At})w$ at small times, whereas (3.20) bounds the displacement of the equilibrium. Both must now be paid, whence the factor two. In the scalar case they coincide to leading order in $\|A_1\|/\lambda$, which is why arbitrary time is available here and unavailable in general. Nor can the threshold be improved by sharpening the estimate: $\|A_1\|/\lambda$ is the exact supremum of the fraction bounded in the proof, approached as $t \to 0$.

The transient of Lemma 3.4 can also be removed when the whole linear part is known and the pair (A, B) is controllable. Recall that the controllability Gramian, written $\mathcal{G}$ so as to leave W to the output matrix,

$$\mathcal{G}(T_0) := \int_0^{T_0} e^{As} BB^\top e^{A^\top s}\, ds \tag{3.41}$$

is then positive definite for every $T_0 > 0$, and that for any target $v \in \mathbb{R}^n$ the control

$$u(t) = B^\top e^{A^\top (T_0 - t)} \mathcal{G}(T_0)^{-1} v \qquad \text{satisfies} \qquad \int_0^{T_0} e^{A(T_0 - s)} Bu(s)\, ds = v; \tag{3.42}$$

see [9, 5].

Lemma 3.7 (Steering lemma in arbitrary time, controllable case). *In the setting of Lemma 3.4, assume moreover that the pair (A, B) is controllable, and write $w := -A^{-1}b$ and $\delta := \delta_K(w)$. Let $T_0 > 0$ and*

$$\mu \;\geq\; \max\left\{1,\; \frac{R + (1 + M)\big(M\rho + M\bar{\phi}/\alpha\big)}{\delta\, \|w\|}\right\}, \tag{3.43}$$

and apply the control

$$u(t) = \begin{cases} \mu\, B^\top e^{A^\top (T_0 - t)} \mathcal{G}(T_0)^{-1} w, & t \in [0, T_0), \\ u_b, & t \geq T_0, \end{cases} \tag{3.44}$$

where $u_b \in \mathbb{R}^m$ is any vector with $Bu_b = \mu b$. Then every solution of (2.12) with $x(0) \in \mathcal{B}(0, \rho)$ satisfies

$$x(t) \in K \quad \text{and} \quad \|x(t)\| \geq R \quad \forall t \geq T_0, \qquad \|x(t)\| \leq \rho' \quad \forall t \geq 0, \tag{3.45}$$

for a radius ρ' given in the proof. This is (3.10), so (2.12) can be steered into K in every time $T_0 > 0$, with no condition on T_0.

Proof. The open-loop part of (3.44) places the state at μw at time T_0, up to the contributions of the initial condition and of ϕ, and the constant part then holds it there. Precisely, by (3.42) we have $\int_0^{T_0} e^{A(T_0 - s)} Bu(s)\, ds = \mu w$, so Duhamel's formula (3.5) at $t = T_0$ gives

$$x(T_0) = \mu w + e^{AT_0} x(0) + \int_0^{T_0} e^{A(T_0 - s)} \phi(s)\, ds, \qquad \big\|x(T_0) - \mu w\big\| \leq M\rho + \frac{M\bar{\phi}}{\alpha}. \tag{3.46}$$

On $[T_0, +\infty)$ the control is the constant $Bu \equiv \mu b$, so Duhamel's formula run from the initial time T_0 and the initial state $x(T_0)$, regrouped as in (3.16), gives

$$x(t) = \mu w + e^{A(t - T_0)}\big(x(T_0) - \mu w\big) + \int_{T_0}^{t} e^{A(t - s)} \phi(s)\, ds, \qquad t \geq T_0, \tag{3.47}$$

whence

$$\big\|x(t) - \mu w\big\| \;\le\; (1+M)\left(M\rho + \frac{M\bar\phi}{\alpha}\right), \qquad t \ge T_0. \tag{3.48}$$

The residual (3.48) carries no term growing with μ, since the target is reached exactly at T_0 instead of asymptotically; this is what removes the minimal time. Since $\mu \ge 1$, (3.3) gives $\operatorname{dist}(\mu w, \mathbb{R}^n \setminus K) \ge \mu\delta\|w\|$, which by (3.43) exceeds the right-hand side of (3.48) by at least R; Proposition 3.2, applied as in the proof of Lemma 3.4 with $r = x(t) - \mu w$, then yields $x(t) \in K$ and $\|x(t)\| \ge R$ for every $t \ge T_0$. On $[0, T_0]$ the state is bounded by $M\rho + M\bar\phi/\alpha + \mu c_B(T_0)\|w\|$, where $c_B(T_0) := \sup_{0\le t\le T_0} \| \int_0^t e^{A(t-s)}BB^\top e^{A^\top(T_0-s)}ds\, \mathcal{G}(T_0)^{-1}\|$ is finite, so (3.45) holds with ρ' the larger of that quantity and $\mu\|w\| + (1+M)(M\rho + M\bar\phi/\alpha)$. ∎

The amplitude (3.43) does not depend on T_0. What grows as T_0 decreases is the control effort, through $\mathcal{G}(T_0)^{-1}$.

3.2. Recursion: generating a saturation-exciting trajectory

Lemma 3.8 (Sequential steering). *Let Assumption 2.8 hold, let A be Hurwitz and, for each $i \in [\![1, n]\!]$, let (2.12) be steerable into K_i in some time $T_0^i > 0$ in the sense of Definition 3.3. Then for every $\rho > 0$ and every*

$$T \;>\; \sum_{i=1}^{n} T_0^i \tag{3.49}$$

there exist pairwise disjoint intervals $J_1, \dots, J_n \subset [0, T]$ of positive length and a control $u : [0, T] \to \mathbb{R}^m$ such that every solution of (2.12) with $x(0) \in \mathcal{B}(0, \rho)$ satisfies

$$x(t) \in K_i \quad \text{and} \quad S(x(t)) = v_i \qquad \forall t \in J_i, \quad \forall i \in [\![1, n]\!]. \tag{3.50}$$

Such a solution realizes the excitation design $\mathcal{E} = \{(v_i, J_i)\}_{i\in[\![1,n]\!]}$ and is therefore saturation-exciting.

Proof. Choose dwell times $\tau_i > 0$ with $\sum_i (T_0^i + \tau_i) \le T$, which is possible since the τ_i may be taken arbitrarily small, and set $s_0 := 0$, $t_i := s_{i-1} + T_0^i$, $s_i := t_i + \tau_i$ and $J_i := [t_i, s_i]$. These intervals are pairwise disjoint and contained in $[0, T]$.

The construction is by induction on i, the induction hypothesis being a bound ρ_i on the norm of the state at time s_{i-1}, with $\rho_1 := \rho$. Since ϕ enters (2.12) only through the time-uniform bound $\|\phi\| \le \bar\phi$, Definition 3.3 may be applied with any initial time. Given ρ_i, it furnishes for K_i and the radius ρ_i a control u_i and a radius ρ_i' such that any solution issued at time s_{i-1} from a state of norm at most ρ_i satisfies $x(t) \in K_i$ and $\|x(t)\| \ge R$ for every $t \ge s_{i-1} + T_0^i$, hence for every $t \in J_i$, and $\|x(t)\| \le \rho_i'$ throughout. By (2.13), $S(x(t)) = v_i$ on J_i, which is (3.50). Setting $\rho_{i+1} := \rho_i'$ closes the induction, and the concatenation $u := \sum_{i=1}^n u_i(\cdot - s_{i-1})\mathbf{1}_{[s_{i-1}, s_i)}$ is the announced control. The J_i are bounded of positive length and contained in $[0, T]$, and $\operatorname{rank}[v_1, \dots, v_n] = n$ by Assumption 2.8, so $\mathcal{E}$ is an excitation design in the sense of Definition 2.2, realized by every such solution. Both the v_i and the J_i are fixed before any solution is considered. ∎

Only the amplitudes depend on the initial radius; the times T_0^i do not, so the induction does not lengthen the horizon as it progresses.

3.3. Proofs of the theorems

Proof of Theorem 2.10. For each i the hypothesis is (3.12) with $K = K_i$; let $\delta_i := \delta^*_{K_i}(A) > 0$. By Lemma 3.4, a direction of depth η in K_i makes (2.12) steerable into K_i in every time $T_0^i > \alpha^{-1}\ln(M/\eta)$, and (3.11) makes every $\eta < \delta_i$ available. Lemma 3.8 then produces the intervals and the control as soon as $T > \sum_i T_0^i$, so the conclusion holds with

$$T^* := \sum_{i=1}^{n} \frac{1}{\alpha} \ln \frac{M}{\delta_i} \ \geq\ 0, \tag{3.51}$$

which is nonnegative because $M \geq 1 \geq \delta_i$. For $T > T^*$ it suffices to take each η close enough to δ_i for the sum to stay below T, which the continuity of the logarithm permits. The deeper the directions available in the K_i, the shorter T^*. $\square$

Proof of Theorem 2.11. For each i let $\delta_i := \delta^*_{K_i}(A_0) > 0$, a quantity determined by A_0, B and K_i alone, and set

$$\varepsilon_0 := \frac{\alpha_0}{M_0} \min_{1\leq i\leq n} \frac{\delta_i}{3+\delta_i}. \tag{3.52}$$

Let $\|A_1\| \leq \varepsilon_0$, so that $(M_0/\alpha_0)\|A_1\| \leq \delta_i/(3+\delta_i)$ for every i. By (3.11) each K_i contains a direction of depth δ_0 as close to δ_i as desired, and since $\delta_i/(3+\delta_i) < \delta_0/(1+\delta_0)$ once δ_0 is close enough to δ_i, such a direction satisfies the hypothesis (3.20) of Lemma 3.5. That lemma therefore applies to each K_i, with $\theta \leq \delta_i/3$, and as $\delta_0 \to \delta_i$ the time in (3.22) decreases to at most

$$T^* := \sum_{i=1}^{n} \frac{2}{\alpha_0} \ln \frac{M_0(3+\delta_i)}{2\delta_i} \ >\ 0, \tag{3.53}$$

because $\theta \leq \delta_i/3$ gives $(\delta_i - \theta)/(1+\theta) \geq 2\delta_i/(3+\delta_i)$. Lemma 3.8 concludes for every $T > T^*$. The directions, the depths δ_i, the times T_0^i and the amplitudes are all computed from A_0, B, the K_i, $\bar\phi$, ρ and the bound ε_0; A_1 itself is never used.

The constant 3 in (3.52), in place of the 1 of (3.29), leaves room for the doubled condition (3.30) of regime R3; case *ii.* of Theorem 2.12 consumes exactly that margin, and the price paid here is a larger T^*. The same ε_0 then serves both theorems. $\square$

Proof of Theorem 2.12, case i.. Here $A_1 = 0$, so $A = A_0$ is known and Hurwitz and the hypothesis of Theorem 2.11 is (3.12) for each K_i. Since (A, B) is controllable, Lemma 3.7 makes (2.12) steerable into K_i in *every* time $T_0^i > 0$. Given $T > 0$, choose $T_0^i := T/(2n)$ and apply Lemma 3.8, whose hypothesis $T > \sum_i T_0^i = T/2$ is satisfied. Hence $T^* = 0$. $\square$

Proof of Theorem 2.12, case ii.. Here $A_0 = -\lambda\,\mathrm{Id}$ with $\lambda > 0$, so $M_0 = 1$, $\alpha_0 = \lambda$ and $A = -\lambda\,\mathrm{Id} + A_1$ has the structure required by Lemma 3.6. By (3.52), $\|A_1\| \leq \varepsilon_0$ gives $\theta \leq \delta_i/3$ for every i, so $2\theta \leq 2\delta_i/3 < \delta_i$ and (3.11) supplies in each K_i a direction of depth $\delta_0 > 2\theta$, which is (3.30). Lemma 3.6 then makes (2.12) steerable into K_i in *every* time $T_0^i > 0$. Given $T > 0$, choose $T_0^i := T/(2n)$ and apply Lemma 3.8, which also returns the design $\mathcal{E}$. Hence $T^* = 0$, the control on the i-th step being computed from λ, B, K_i, $\bar\phi$, ρ and the bound ε_0. $\square$

4. Proof of Theorem 2.15 and Corollary 2.16

4.1. The output feedback law

The mechanism of this section is not the one of Section 3: no direction is aimed at, and what plays the role of the amplitude is a scalar gain k, written so as to leave

μ to the steering. The deadband ω below is measured on the output side and is unrelated to the margin δ_K of Definition 3.1. For $k > 0$, $\omega \in (0, 1/2)$ and a reference value $r \in \mathbb{R}$, define the scalar feedback

$$\eta_0(\sigma; k, r, \omega) = \begin{cases} k, & \sigma \le 0, \\ \ell_1(\sigma; k, r), & 0 < \sigma < \omega, \\ -k\bigl(S_0^{-1}(\sigma) - r\bigr), & \omega \le \sigma \le 1 - \omega, \\ \ell_2(\sigma; k, r), & 1 - \omega < \sigma < 1, \\ -k, & \sigma \ge 1, \end{cases} \tag{4.1}$$

where ℓ_1 and ℓ_2 are the functions of σ, affine on $[0, \omega]$ and on $[1 - \omega, 1]$ respectively, making η_0 continuous. The law pushes the state back when the measurement indicates saturation, and switches to proportional tracking once the measurement lies in the invertible range of S_0. The feedback of Theorem 2.15 is obtained by applying (4.1) to each coordinate of $Hy = S_1(x)$, with the corresponding coordinate of the reference, and inverting B: writing, for $\sigma \in \mathbb{R}^n$ and $r \in \mathbb{R}^n$,

$$\boldsymbol{\eta}_0(\sigma; k, r, \omega) := \bigl(\eta_0(\sigma_1; k, r_1, \omega), \dots, \eta_0(\sigma_n; k, r_n, \omega)\bigr)^\top, \tag{4.2}$$

the feedback is

$$\eta(y, \gamma) := B^{-1} \boldsymbol{\eta}_0\bigl(Hy; k, \gamma, \omega\bigr). \tag{4.3}$$

Since B^{-1} is applied last, the control enters the dynamics as

$$Bu = \boldsymbol{\eta}_0\bigl(S_1(x); k, \gamma, \omega\bigr), \tag{4.4}$$

so that the loop is closed coordinate by coordinate and the analysis is scalar.

Two objects are needed for the statement. For $\omega \in (0, 1/2)$ write

$$\Omega_\omega := \bigl[S_0^{-1}(\omega),\ S_0^{-1}(1 - \omega)\bigr]^n \subset \mathring{\Omega} \tag{4.5}$$

for the box obtained from Ω by staying at distance ω from the saturation levels, measured on the output side, and, for $L > 0$ and $X \subset \mathbb{R}^n$, let $C^{1,L}(\mathbb{R}, X)$ denote the set of C^1 curves with values in X whose derivative is bounded by L.

Lemma 4.1 (Tracking under a saturated measurement). *Let B be invertible, let $g : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R} \to \mathbb{R}^n$ be continuous with*

$$\|g(x, u, t)\| \le a\|x\| + c_g, \qquad (x, u, t) \in \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}, \tag{4.6}$$

for some known *constants $a, c_g > 0$, and consider*

$$\dot{x} = g(x, u, t) + Bu, \qquad y = WS(x), \tag{4.7}$$

with W known and H, S_0, S_1 as in Assumption 2.13. Let $\varepsilon, \rho, L > 0$, let $0 < T_1 < T_2$, let $\omega \in (0, 1/4)$ and let $\gamma \in C^{1,L}(\mathbb{R}, \Omega_{2\omega})$. Set

$$\Lambda := \max\bigl\{\rho, |\underline{R}|, |\bar{R}|\bigr\}, \qquad \kappa := a\sqrt{n}\,\Lambda + c_g + L, \tag{4.8}$$

$$c_\omega := \min\Bigl\{1,\ S_0^{-1}(1 - \omega) - S_0^{-1}(1 - 2\omega),\ S_0^{-1}(2\omega) - S_0^{-1}(\omega)\Bigr\}\ >\ 0. \tag{4.9}$$

Then, for every gain

$$k \ > \ k^\star := \max\left\{ \frac{1}{c_\omega}\left(\frac{2\Lambda}{T_1} + a\sqrt{n}\,(\Lambda+1) + c_g\right),\quad \frac{2\sqrt{n}\,\kappa}{\varepsilon},\quad \frac{1}{T_2 - T_1}\ln\frac{2\sqrt{n}\,(\bar R - \underline{R})}{\varepsilon}\right\}, \tag{4.10}$$

the solutions of (4.7) *under the control* $u(t) = \eta(y(t), \gamma(t))$ *of* (4.3)*, issued from* $\mathcal{B}(0,\rho)$*, are global in time and verify*

$$x(t) \in \Omega_\omega \quad \forall t \geq T_1, \qquad \|x(t) - \gamma(t)\| \leq \varepsilon \quad \forall t \geq T_2. \tag{4.11}$$

The three terms of (4.10) are, in order, what makes Ω_ω attractive before T_1 and invariant afterwards, what bounds the asymptotic tracking offset by ε, and what makes the initial error decay to ε before T_2. All of them are computed from declared data: the gain requires the constants a and c_g of (4.6), so an upper bound on the growth of g has to be known, even though g itself need not be.

The reference is asked to keep the deadband 2ω while the feedback is built with the deadband ω. The gap makes the proportional term nonzero at the edge of Ω_ω, hence makes that box invariant; with a single deadband the term vanishes when γ sits exactly on the edge, and nothing pushes the state back.

Proof. By (2.15), $Hy(t) = HWS(x(t)) = S_1(x(t))$, so (4.3) is indeed an output feedback. Continuity of S, g and η_0 gives local existence and absolute continuity of the solutions, and as noted after (4.3) the closed loop reads

$$\dot x_j = g_j(x,u,t) + \eta_0\big(S_0(x_j); k, \gamma_j, \omega\big), \qquad j \in [\![1,n]\!]. \tag{4.12}$$

The positivity of c_ω in (4.9) follows from $\omega < 1/4$ and the strict monotonicity of S_0 on $[\underline{R}, \bar R]$. Throughout, k satisfies (4.10), whose first term in particular gives

$$k \ > \ \frac{a\sqrt{n}\,(\Lambda+1) + c_g}{c_\omega}. \tag{4.13}$$

Step 1: the feedback is uniformly negative on the upper saturated range. Since $\gamma_j(t) \in [S_0^{-1}(2\omega), S_0^{-1}(1-2\omega)]$, the value of η_0 at $\sigma = 1-\omega$ is $-k(S_0^{-1}(1-\omega) - \gamma_j) \leq -kc_\omega$ by (4.9), and its value at $\sigma = 1$ is $-k \leq -kc_\omega$. Being affine in σ on $[1-\omega, 1]$ and constant beyond, η_0 is therefore at most $-kc_\omega$ on the whole of $[1-\omega, +\infty)$. Symmetrically $\eta_0 \geq kc_\omega$ on $(-\infty, \omega]$. In terms of the state,

$$\eta_0\big(S_0(\xi); k, \gamma_j, \omega\big) \begin{cases} \leq -kc_\omega, & \xi \geq S_0^{-1}(1-\omega), \\ \geq \ \ kc_\omega, & \xi \leq S_0^{-1}(\omega). \end{cases} \tag{4.14}$$

Step 2: an invariant box, and globality. We claim that $\{\|x\|_\infty \leq \Lambda\}$ is positively invariant, and argue on the slightly larger box $\{\|x\|_\infty \leq \Lambda+1\}$, on which (4.6) still yields a usable bound. Suppose the solution leaves $\{\|x\|_\infty \leq \Lambda\}$. As $t \mapsto \|x(t)\|_\infty$ is continuous and $\|x(0)\|_\infty \leq \|x(0)\| \leq \rho \leq \Lambda$, it reaches the value $\Lambda+1$; let t_1 be the first time at which it does and let j be an index with $|x_j(t_1)| = \Lambda+1$, say $x_j(t_1) = \Lambda+1$, the other case being symmetric. Since $x_j(0) \leq \Lambda$, the set of times before t_1 at which $x_j = \Lambda$ is nonempty; let t_0 be its supremum, so that $x_j > \Lambda$ on $(t_0, t_1]$. On $[t_0, t_1]$ we have $\|x\|_\infty \leq \Lambda+1$, because t_1 is the first time that value is reached, so (4.6) gives $|g_j| \leq a\sqrt{n}(\Lambda+1) + c_g$; and $x_j \geq \Lambda \geq \bar R \geq S_0^{-1}(1-\omega)$ there, so (4.12) and (4.14) give

$$\dot x_j \ \leq \ a\sqrt{n}\,(\Lambda+1) + c_g - kc_\omega \ < \ 0 \tag{4.15}$$

almost everywhere on $[t_0, t_1]$ by (4.13), contradicting $x_j(t_1) > x_j(t_0)$. The solution therefore stays in the box, hence is global, and

$$\|g(x(t), u(t), t)\| \ \leq \ a\sqrt{n}\,\Lambda + c_g, \qquad \forall t \geq 0. \tag{4.16}$$

Step 3: Ω_ω is reached before T_1 and is invariant. By (4.14) and (4.16), whenever $x_j(t) \geq S_0^{-1}(1-\omega)$ one has $\dot{x}_j \leq -(kc_\omega - a\sqrt{n}\Lambda - c_g) < 0$, and symmetrically $\dot{x}_j > 0$ whenever $x_j(t) \leq S_0^{-1}(\omega)$. The interval $[S_0^{-1}(\omega), S_0^{-1}(1-\omega)]$ is therefore positively invariant for x_j, and since $|x_j(0)| \leq \Lambda$ and $|S_0^{-1}(\sigma)| \leq \Lambda$ for $\sigma \in [0,1]$, it is reached after a time at most

$$\frac{2\Lambda}{kc_\omega - a\sqrt{n}\,\Lambda - c_g}, \tag{4.17}$$

which tends to 0 as $k \to \infty$ and is at most T_1 by the first term of (4.10). Hence $x(t) \in \Omega_\omega$ for every $t \geq T_1$, the first assertion of (4.11).

Step 4: tracking. For $t \geq T_1$ we have $S_0(x_j(t)) \in [\omega, 1-\omega]$, so $S_0^{-1}(S_0(x_j)) = x_j$ and (4.12) reads $\dot{x}_j = g_j - k(x_j - \gamma_j)$. With $e_j := x_j - \gamma_j$ and $\|\dot{\gamma}\| \leq L$, (4.16) and (4.8) give

$$D^+|e_j| \ \leq \ \kappa - \mu|e_j| \qquad \text{almost everywhere on } [T_1, +\infty), \tag{4.18}$$

where D^+ is the upper right Dini derivative, $|e_j|$ being absolutely continuous but not differentiable at its zeros. Whence, by Grönwall's inequality and $|e_j(T_1)| \leq \bar{R} - \underline{R}$,

$$|e_j(t)| \ \leq \ \big(\bar{R} - \underline{R}\big)e^{-\mu(t-T_1)} + \frac{\kappa}{\mu}, \qquad t \geq T_1. \tag{4.19}$$

The constant κ does not depend on μ, so both terms are at most $\varepsilon/(2\sqrt{n})$ for every $t \geq T_2$ as soon as

$$k \ > \ \max\left\{\frac{2\sqrt{n}\,\kappa}{\varepsilon},\ \frac{1}{T_2 - T_1}\ln\frac{2\sqrt{n}\,(\bar{R} - \underline{R})}{\varepsilon}\right\}, \tag{4.20}$$

which are the two remaining terms of (4.10). Then $\|x(t) - \gamma(t)\| \leq \sqrt{n}\max_j |e_j(t)| \leq \varepsilon$ for $t \geq T_2$. ∎

Remark 4.2 (Practical, not asymptotic, tracking). The bound (4.19) does not give $x(t) - \gamma(t) \to 0$, and no such conclusion is available here. The field g is only known through the bound (4.6), and against a constant disturbance a proportional law leaves a constant offset: if $g \equiv g_0$ and γ is constant, the loop (4.12) has the equilibrium $\gamma_j + g_{0,j}/k$ and $|e_j(t)| \to |g_{0,j}|/k$. The content of (4.19) is that this residual offset is at most κ/k, so that any prescribed accuracy is reached, at a prescribed time, by taking the gain large enough. Asymptotic tracking would require integral action, hence a dynamic feedback and enough knowledge of g to place it.

Only one property of the output was used, namely that Hy be a component-wise saturation of the state. Outside the class (2.15) the scalar law (4.1) has no component-wise meaning: the unsaturated region is no longer a box and its faces are not aligned with the coordinates, so the recovery below does not carry over. The steering results of Section 3 hold for general coradiant regions; only this last stage is tied to the orthant geometry.

Proof of Theorem 2.15. Let $\mathcal{K} \subset \mathring{\Omega}$ be a compact set containing the values of γ and let $L > 0$ bound $\|\dot\gamma\|$. Since $\mathcal{K}$ is compact and contained in the open box $\mathring{\Omega}$, there is $\zeta > 0$ with $\mathcal{K} \subset [\underline{R}+\zeta, \bar R - \zeta]^n$, and then, S_0^{-1} being continuous on $[0,1]$ with $S_0^{-1}(0) = \underline{R}$ and $S_0^{-1}(1) = \bar R$, some $\omega \in (0, 1/4)$ with $\mathcal{K} \subset \Omega_{2\omega}$; thus $\gamma \in C^{1,L}(\mathbb{R}, \Omega_{2\omega})$. The system (2.12) is of the form (4.7) with $g(x,u,t) = Ax + \phi(x,u,t)$, which is continuous and satisfies (4.6) with $a = \|A\| > 0$ and $c_g = \bar\phi > 0$, uniformly in u and t. Lemma 4.1 then gives (4.11), and $\Omega_\omega \subset \mathring{\Omega}$ turns its first assertion into that of (2.17). The deadband ω and the gain threshold $k^\star$ of (4.10) are the explicit output of this construction; note that $k^\star$ requires known upper bounds on $\|A\|$ and on $\bar\phi$, whereas A and ϕ themselves remain unknown. ◻

4.2. Recovering the state from the output

It remains to invert the output map on the unsaturated region. Two inversions are involved, of different natures: that of W, which is linear and holds at all times, and that of the saturation, which is nonlinear and holds only where the state is unsaturated.

Proof of Corollary 2.16. By (2.15) the signal $Hy(t) = S_1(x(t))$ is available at every time and whatever the state: the output matrix destroys no information, the saturation does. Next, S_0 is continuous and strictly increasing on $[\underline{R}, \bar R]$ with $S_0(\underline{R}) = 0$ and $S_0(\bar R) = 1$, hence an increasing bijection from $[\underline{R}, \bar R]$ onto $[0,1]$, whose inverse is continuous by compactness. Applying this coordinate by coordinate, the restriction of S_1 to Ω is a homeomorphism onto $[0,1]^n$, with inverse $(\sigma_1, \dots, \sigma_n) \mapsto (S_0^{-1}(\sigma_1), \dots, S_0^{-1}(\sigma_n))$. By Theorem 2.15 the state lies in $\mathring{\Omega}$ for $t \ge T_1$, and (2.18) follows by applying that inverse to $Hy(t) = S_1(x(t))$. ◻

Theorem 2.15 delivers more than $x(t) \in \mathring{\Omega}$: it keeps the state at distance ω from the saturation levels. The next remark quantifies what that margin buys on measured data.

Remark 4.3 (Conditioning of the recovery). Assume S_0 is C^1 with $S_0' > 0$ on $(\underline{R}, \bar R)$ and set $m_\omega := \min\{S_0'(\xi) : S_0(\xi) \in [\omega/2, 1-\omega/2]\} > 0$, where ω is the deadband produced by the proof of Theorem 2.15. Suppose the output is measured as $y+\nu$ with $\|H\nu\|_\infty \le \min\{\bar\nu, \omega/2\}$. For $t \ge T_1$ the state lies in Ω_ω, so the coordinates of $H(y+\nu)$ still lie in $[\omega/2, 1-\omega/2]$, on which S_0^{-1} is m_ω^{-1}-Lipschitz, and the recovered state $\hat x_j := S_0^{-1}\big((H(y+\nu))_j\big)$ obeys

$$\|\hat x(t) - x(t)\|_\infty \;\le\; \frac{\bar\nu}{m_\omega}, \qquad t \ge T_1. \tag{4.21}$$

For a saturation which is C^1 across its levels, S_0' vanishes there and $m_\omega \to 0$ as $\omega \to 0$: the recovery is ill-conditioned near saturation. This, rather than the mere invertibility of S_0, is why the feedback is asked to keep the state at a positive distance from the levels. The noise is accounted for here in the recovery step alone, the trajectory being the one produced by the noiseless loop of Theorem 2.15. If the feedback is itself driven by $y+\nu$, the sign estimate (4.14) survives with c_ω replaced by $c_\omega - \|H\nu\|_\infty/m_\omega$, so the same conclusions hold for a noise small in that ratio, at the price of a larger gain; we do not pursue this.

5. Numerical simulation

We illustrate the identification of W on a two-dimensional example, the smallest one in which the mechanism is visible: two regions have to be visited, and the excitation design is read directly on the phase plane. We take $n = q = 2$, $m = 1$ and consider

$$
\text{(5.1)} \qquad \begin{cases} \dot{x} = -\lambda x + CS(x) + w_k(t) + Bu(t), \\ y = WS(x) + \nu_k(t), \end{cases}
$$

with $B = \begin{bmatrix} 1 & -1 \end{bmatrix}^\top$, $\lambda = 1.2$, and C, W unknown matrices whose entries are drawn independently and uniformly in $[-1, 1]$. The system is integrated by the Euler–Maruyama scheme with step $T_s = 10^{-3}$, and w_k and ν_k are independent zero-mean Gaussian variables drawn afresh at each step, with standard deviations $\sigma_w = 0.5$ and $\sigma_\nu = 0.3$; the output noise is thus a discrete-time measurement noise attached to the sampling period T_s, and the errors reported below scale like $\sigma_\nu (T_s/\tau)^{1/2}$; the sampling period is therefore part of the experiment rather than an artefact of the integrator. The saturation is piecewise linear with levels $\underline{R} = 0$ and $\bar{R} = 1.5$,

$$
\text{(5.2)} \qquad S_0(\xi) = \operatorname{clamp}\left(\frac{\xi}{1.5}, 0, 1\right) = \begin{cases} 0 & \xi \le 0, \\ \dfrac{\xi}{1.5} & 0 < \xi < 1.5, \\ 1 & \xi \ge 1.5, \end{cases}
$$

applied component-wise. In the notation of Subsection 2.2 this is (2.12) with $A = -\lambda\,\mathrm{Id}$, that is $A_0 = -\lambda\,\mathrm{Id}$ and $A_1 = 0$, and with $\phi = CS(x) + w_k$. The coupling term $CS(x)$ is bounded by $\|C\|$, since $\|S\|_\infty = 1$; the Gaussian process w_k is not, so the simulation probes the method slightly outside the class covered by the theorems, which assume a uniform bound $\bar{\phi}$ on the whole of ϕ.

The image of B is the line spanned by $(1, -1)^\top$, which meets the two orthants of pattern $(+, -)$ and $(-, +)$; the corresponding saturation values are $v_1 = (1, 0)^\top$ and $v_2 = (0, 1)^\top$, which are linearly independent, so Assumption 2.8 is satisfied with the two sets (2.14)

$$
K_1 = [1.5, +\infty) \times (-\infty, -1.5], \qquad K_2 = (-\infty, -1.5] \times [1.5, +\infty),
$$

which correspond to $R = \max\{|\underline{R}|, |\bar{R}|\} = 1.5$. Since $A_1 = 0$, case *ii.* of Theorem 2.12 applies and the design may be realized on an arbitrarily short horizon; we take the windows long instead, $\tau = 10$, in order to display the averaging effect of (2.6) on the output noise. With $t_1 = \tau$ and $t_2 = t_1 + 2\tau$, the piecewise constant control is $u = \mu_1 \mathbf{1}_{[0, t_1+\tau]} - \mu_2 \mathbf{1}_{[t_1+\tau, t_2+\tau]}$ with $\mu_1 = 10$ and $\mu_2 = 12$, which is the control of Lemma 3.8 for this design; the associated equilibria are $\mu_1(1, -1)^\top/\lambda \simeq (8.3, -8.3)$ and $\mu_2(-1, 1)^\top/\lambda \simeq (-10, 10)$, both well inside the respective regions.

The matrix W is then estimated by the estimator $E(y; \mathcal{E})$ of (2.6) with $\mathcal{E} = \{(v_1, [t_1, t_1+\tau]), (v_2, [t_2, t_2+\tau])\}$, the integrals being evaluated by the trapezoidal rule on the sampled output. Over 100 runs, with C, W and the noises redrawn each time, the error $\|E(y; \mathcal{E}) - W\|$ had mean 9×10^{-3}. Two complementary mechanisms account for this. During the saturated phases $S(x(t)) = v_i$ regardless of small perturbations of x, so the estimator is insensitive to the state noise w_k; and the output noise ν_k being zero-mean, the time averaging in (2.6) acts as a low-pass filter and reduces its contribution by a factor of order $\tau^{-1/2}$.

Figure 5.1 shows the trajectory in the phase plane, with the two segments on which $S(x(t)) = v_i$ highlighted; these are the segments the estimator integrates over.

6. Conclusion

The obstruction addressed here is a product of two unknowns: the parameters to be identified multiply a signal which is itself unknown, produced by unknown dynamics

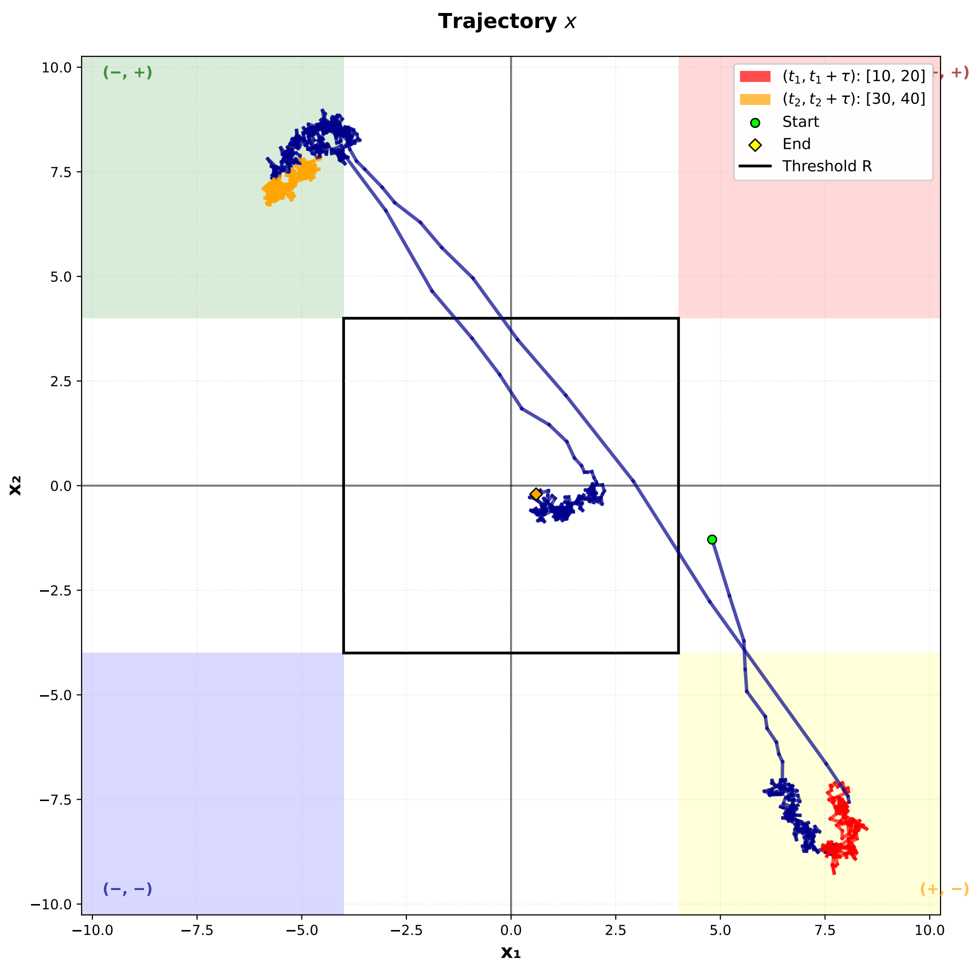


FIGURE 5.1. *Trajectory of x in the phase plane, colour-coded by saturation regime. The red segment is the interval $[t_1, t_1 + \tau]$, on which $S(x(t)) = v_1 = (1, 0)^\top$, and the yellow segment is $[t_2, t_2 + \tau]$, on which $S(x(t)) = v_2 = (0, 1)^\top$; the grey portions are the transients between them. The two highlighted segments carry the two independent values that the estimator* (2.6) *needs to recover the whole of W.*

and never measured. No classical excitation condition helps, since every such condition bears on the regressor, and here the regressor is precisely what nobody sees. The saturation breaks the product. On the regions where S plateaus, an unmeasured state produces a value that is known in advance, so the regressor stops being something to be estimated and becomes something to be *manufactured*: the control designs it, and the estimator then reads W off a linear regression with an exactly known regressor. Identifiability accordingly rests no longer on a persistency property of a trajectory nobody observes, but on the position of $\mathrm{Im}(A^{-1}B)$ relative to the saturation regions, a condition on the data, decidable offline. The feature usually treated as the obstruction, namely that saturation destroys information about the state, is here the resource.

The price is control authority. The amplitudes μ of Section 3 are not small, and the state must be driven deep into saturation and held there; ϕ must be uniformly bounded, and the gain of Section 4 requires known upper bounds on $\|A\|$ and on $\bar{\phi}$, while B must be invertible for the recovery stage. Tracking is practical, not asymptotic. Noise is treated in the recovery step only, in Remark 4.3, and empirically for the identification step; a robustness statement for the estimator, and the infinite-dimensional case, are the natural next steps.

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